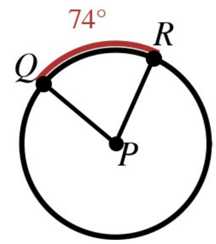
## **S**peak **L**ike **A** **M**athematician:




congruent chords:

distance from center to a chord:

## Chord Theorems

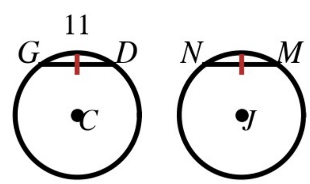
**In the same circle (or congruent circles), two chords are congruent if and only if their ___________________ are congruent.**

**If a diameter is perpendicular to a chord, it ___________________ the chord AND its arc.**

**Two chords are congruent if and only if they are ___________________ from the center.**

## Example 1 – Central Angle and Arc

a) In circle P (top right), the central angle measures 74°. Find m$\overset{\frown}{QR}$.




## Example 2 – Congruent Circles

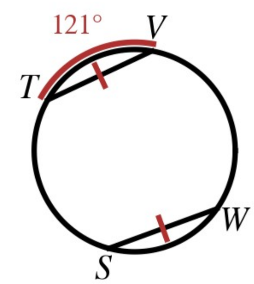
Circles C and J are congruent, and chords GD and NM each have length 11.

a) What can you conclude about arcs GD and NM? Why?

## Example 3 – Same Circle

In the circle at the right, chords TV and WS are marked congruent, and $m\widehat{TV} = 121°$.

a) Find $m\widehat{WS}$.

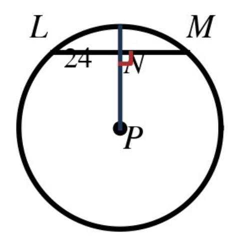



## Example 4 – Perpendicular Bisector in Action

In circle P, a radius is perpendicular to chord LM at N, and LM = 24.

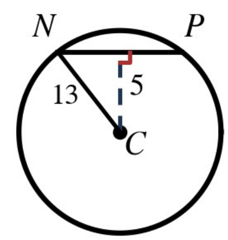
a) Find LN and NM.

b) What else gets bisected?



## Example 5 – Radius, Chord, Distance

In circle C at the right, the radius CN = 13 and the center is 5 units from chord NP.




a) Find NP.

## Example 6 – Find the Radius

a) A chord of length 40 is 15 units from the center. Find the radius.

## Example 7 – Equidistant Chords

Chords AB = 2x + 5 and CD = x + 11 are the same distance from the center.

a) Find x.

b) Find the length of each chord.